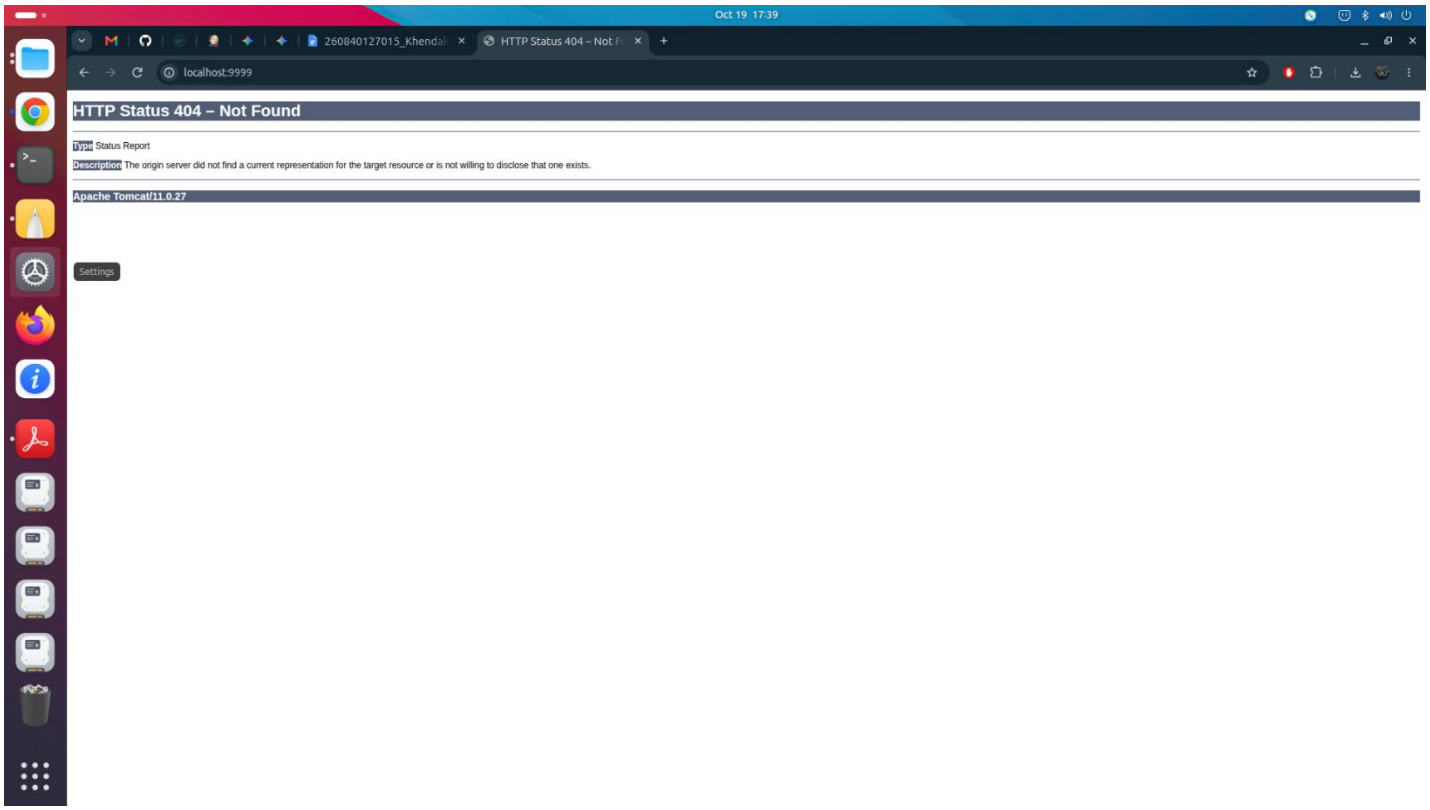
Assignment 3 (Cloud Services & Security)

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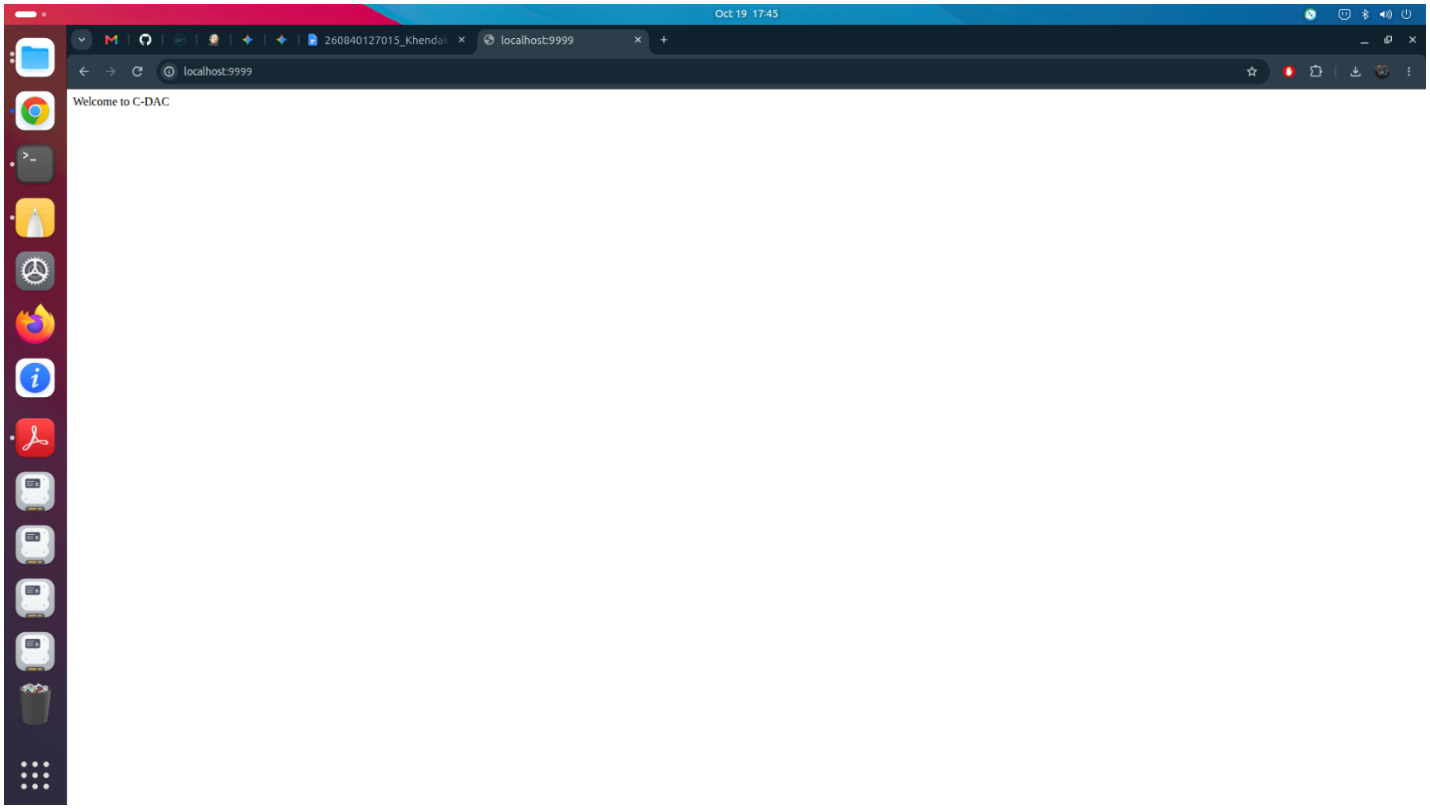
1. Run tomcat web server and access it on local port 9999

```
pradip@ProBook:~/docker_jenkins_demo$ cd
pradip@ProBook:~$ docker run -d --name tomcat-server -p 9999:8080 tomcat:latest
Unable to find image 'tomcat:latest' locally
latest: Pulling from library/tomcat
4f4fb700ef54: Pull complete
fc48198bd247: Pull complete
187036d87529: Pull complete
435bf5c20694: Already exists
d9d511d19a19: Already exists
5ae6b7a60007: Pull complete
2f8182fc09a5: Pull complete
fdf74691d629: Pull complete
3f3033de63da: Download complete
4688fd026291: Download complete
Digest: sha256:fc6089642155a7cdad0cf118cf1fcca74a2f89e90169bdf913b2e38e603c0b47
Status: Downloaded newer image for tomcat:latest
0431790eea9d2b68ba7e2240b0f969c6130c971213fdcd81e180d5f6d8b5add4
```

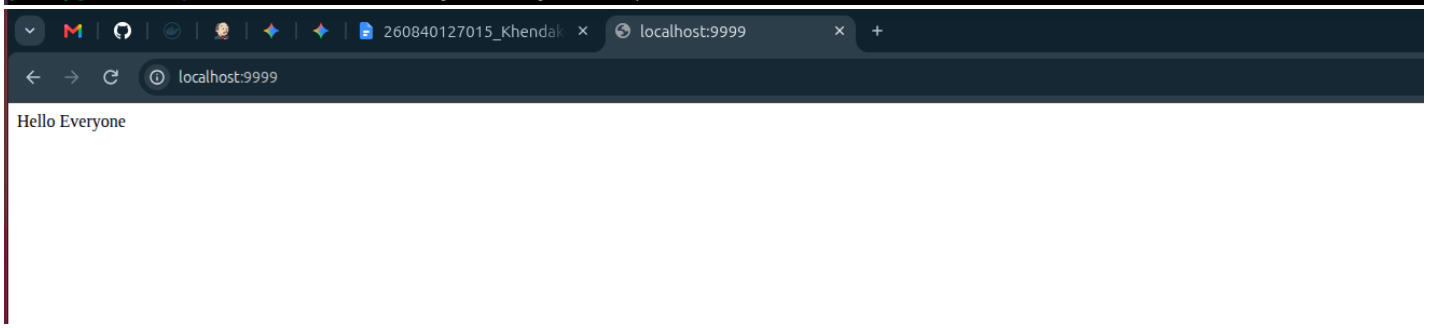


2. Run apache web server and access it on local port 9999. Use Docker volume to change the home page to display "Welcome to C-DAC". Verify it is visible in the browser. Then change it to "Hello Everyone". Verify again in the browser.

```
pradip@ProBook:~$ mkdir assi4
pradip@ProBook:~$ cd assi4/
pradip@ProBook:~/assi4$ docker run -d --name tomcat-server -p 9999:8080 tomcat:latest
d8b09cf0d36559366fa2fc294e86938241586a063b034314694ac32b7220654e
pradip@ProBook:~/assi4$ docker stop tomcat-server
tomcat-server
pradip@ProBook:~/assi4$ docker rm tomcat-server
tomcat-server
pradip@ProBook:~/assi4$ mkdir -p my-website
pradip@ProBook:~/assi4$ echo "Welcome to C-DAC" > my-website/index.html
pradip@ProBook:~/assi4$ docker run -d --name apache-server -p 9999:80 -v $(pwd)/my-website:/usr/local/apache2/htdocs/ httpd:latest
190876938b6fde6adc9a88337b99f38f42b3cbc3258c3d29fb6b4f8c9a4c737c
```




```
pradip@ProBook:~/assi4$ echo "Hello Everyone" > my-website/index.html
```



3. Run a Python program that adds your name and phone number to a MySQL table named "mydata". Run these two in two separate Docker containers. Network them together and verify that the data exists by opening a terminal inside MySQL container.

Create a Docker Network-

```
pradip@ProBook:~/assi4$ docker network create my-net
23bbd1c6c6bf2d42e458b9432f8befc847278993f7acee6ba1bf8ce62b6db8ef
```

Run MySQL Container-

```
pradip@ProBook:~/assi4$ docker run -d --name mysql-db   --network my-net   -e MYSQL_ROOT_PASSWORD=rootpassword   -e MYSQL_DATABASE=testdb   mysql:latest
9c80c49eaec784c3a0fa073c1d50c2fa5114ddef3622a691c39d9400e3103825
```

Create the Python Script & Run Container

```
pradip@ProBook:~/assi4$ nano app.py
```

```
 GNU nano 7.2                                  app.py *
import mysql.connector

db = mysql.connector.connect(
    host="mysql-db",
    user="root",
    password="rootpassword",
    database="testdb"
)

cursor = db.cursor()
cursor.execute("CREATE TABLE IF NOT EXISTS mydata (name VARCHAR(255), phone VARCHAR(255))")

cursor.execute("INSERT INTO mydata (name, phone) VALUES ('John Doe', '1234567890')")
db.commit()

print("Data inserted successfully!")
db.close()
```

Run a Python container, install the MySQL connector, and executing script:

```
pradip@ProBook:~/assi4$ docker run -it --rm --name python-app --network my-net -v $(pwd):/app -w /app python:3.9-slim bash
root@71e4c7e36ad5:/app# pip install mysql-connector-python
Collecting mysql-connector-python
  Downloading mysql_connector_python-9.4.0-cp39-cp39-manylinux_2_28_x86_64.whl (33.9 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 33.9/33.9 MB 3.8 MB/s eta 0:00:00
Installing collected packages: mysql-connector-python
Successfully installed mysql-connector-python-9.4.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. I
stead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
root@71e4c7e36ad5:/app# python app.py
Data inserted successfully!
root@71e4c7e36ad5:/app# exit
exit
pradip@ProBook:~/assi4$ docker exec -it mysql-db mysql -u root -prootpassword testdb -e "SELECT * FROM mydata;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+----------+------------+
| name     | phone      |
+----------+------------+
| John Doe | 1234567890 |
+----------+------------+
```

4. Do the above. This time also attach a volume to the MySQL container. Rerun the application. Remove MySQL container. Verify that the inserted data still exists.

```
pradip@ProBook:~/assi4$ docker run -d --name mysql-db --network my-net -v mysql_volume:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=rootpassword -e MYSQL_DATABASE=testdb mysql:latest
6044559cc341d73a0b26049c7a9955fe1d7797d953920fbe859da792a038e862
pradip@ProBook:~/assi4$ docker run -it --rm --name python-app --network my-net -v $(pwd):/app -w /app python:3.9-slim bash
root@454b2c39a437:/app# pip install mysql-connector-python
Collecting mysql-connector-python
  Downloading mysql_connector_python-9.4.0-cp39-cp39-manylinux_2_28_x86_64.whl (33.9 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 33.9/33.9 MB 2.5 MB/s eta 0:00:00
Installing collected packages: mysql-connector-python
Successfully installed mysql-connector-python-9.4.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment i
stead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
root@454b2c39a437:/app# python app.py
Data inserted successfully!
root@454b2c39a437:/app# exit
exit
pradip@ProBook:~/assi4$ docker rm -f mysql-db
mysql-db
pradip@ProBook:~/assi4$ docker run -d --name mysql-db --network my-net -v mysql_volume:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=rootpassword -e MYSQL_DATABASE=testdb mysql:latest
a4e5a2a299fb48528fd2c8f4a2c560090f62257401c1a80cbba46c28e41bb0b5
pradip@ProBook:~/assi4$ docker exec -it mysql-db mysql -u root -prootpassword testdb -e "SELECT * FROM mydata;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+----------+------------+
| name     | phone      |
+----------+------------+
| John Doe | 1234567890 |
+----------+------------+
```

5. Create a hello python application using Docker Compose. Verify its output.

```
pradip@ProBook:~/assi4$ nano Dockerfile
pradip@ProBook:~/assi4$ nano main.py
pradip@ProBook:~/assi4$
pradip@ProBook:~/assi4$ nano docker-compose.yml
pradip@ProBook:~/assi4$ docker compose up --build
WARN[0000] /home/pradip/assi4/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 13.3s (10/10) FINISHED
 => [internal] load local bake definitions                                                                                                                                0.0s
 => => reading from stdin 504B                                                                                                                                            0.0s
 => [internal] load build definition from Dockerfile                                                                                                                      0.4s
 => => transferring dockerfile: 109B                                                                                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                                                                                        0.2s
 => [internal] load .dockerignore                                                                                                                                         0.3s
 => => transferring context: 2B                                                                                                                                           0.0s
 => [1/3] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b                                                  3.5s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b                                                  0.5s
 => => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d                                                                                 0.8s
 => => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d                                                                                 0.3s
 => => extracting sha256:fc7443084902 2d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08                                                                                0.6s
 => => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508                                                                                 0.2s
 => [internal] load build context                                                                                                                                         0.6s
 => => transferring context: 878B                                                                                                                                         0.0s
 => [2/3] WORKDIR /app                                                                                                                                                    3.7s
 => [3/3] COPY . /app                                                                                                                                                     0.6s
 => exporting to image                                                                                                                                                    2.9s
 => => exporting layers                                                                                                                                                   1.6s
 => => exporting manifest sha256:0b33d66f9f0d88bc729eac782c2debcf99001b2bdf73ca7b9e589bc3ba024ff1                                                                         0.1s
 => => exporting config sha256:b0a4554a0b0716a8d780ecd4574f4f9120b92c75db57ea8727cf7d98b844ab72                                                                           0.1s
 => => exporting attestation manifest sha256:d24e2e438d9826afbc818ef23befac173bd48fbc65871f8c1c0e1a5519b56057                                                             0.2s
 => => exporting manifest list sha256:877205dce113d1722bb59f4ca2d4f383d6bae9faeea0524bbe86c3508e658f1c                                                                    0.2s
 => => naming to docker.io/library/assi4-hellopython:latest                                                                                                               0.0s
 => => unpacking to docker.io/library/assi4-hellopython:latest                                                                                                            0.5s
 => resolving provenance for metadata file                                                                                                                                0.0s
[+] up 3/3
 ✔ Image assi4-hellopython      Built                                                                                                                                    13.1s
 ✔ Network assi4_default        Created                                                                                                                                   0.4s
 ✔ Container compose_python_app Created                                                                                                                                   1.1s
Attaching to compose_python_app
compose_python_app  | Hello from Docker Compose Python Application!
compose_python_app exited with code 0
```